

CYBER SECURITY INTRODUCTION

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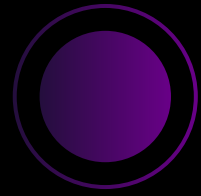
Get Started



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Our Trainer



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Origins



Rise of Sophisticated Attacks
(2000s)

Operation Aurora (2009)

Cybercrime Becomes
Organized (2010s)

Stuxnet (2010)

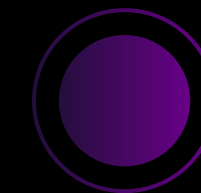
WannaCry in 2017



Early Computing and First Cyber
Attacks (1960s–1980s)

Creeper virus

Morris Worm



Why Cybersecurity?

Increased Threat Surface

5G Networks

Cloud computing

IoT

Assuring Business Continuity

Supply chains

Critical Infrastructure

DATA PROTECTION

Cloud security vulnerabilities

Nation-state attacks

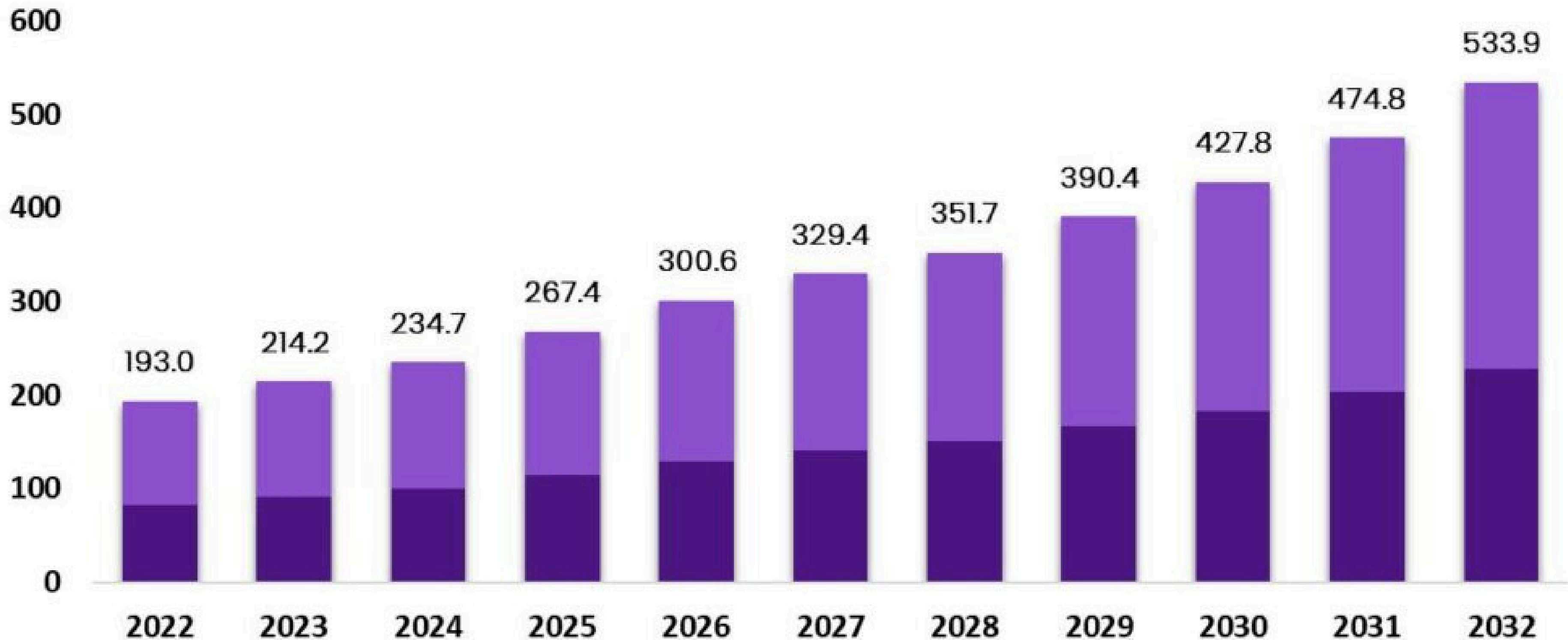


Global Cyber Security Market

Size, by Component Type, 2022-2032 (USD Billion)

Solutions

Services



The Market will Grow
At the CAGR of:

11%

The forecasted market
size for 2032 in USD:

\$533.9B

 **market.us**
ONE STOP SHOP FOR THE REPORTS

Key Fields in Defensive security

Data Protection – Safeguarding sensitive data from unauthorized access and breaches.

Risk Management – Identifying, evaluating and mitigating potential security risks.

Network Security – Protecting and monitoring network traffic to prevent unauthorized access.



Key Fields in Offensive security



Red Teaming – Conducting full-scope attack simulations to test defenses and improve security.

Penetration Testing –
Simulating cyberattacks to
identify vulnerabilities in
systems.

Exploit Development

Getting Started In Cybersecurity



- **The Basics**

Networking Fundamentals: TCP/IP, DNS, Firewalls, VPNs..etc

Operating Systems: Windows, Linux, and their security features.

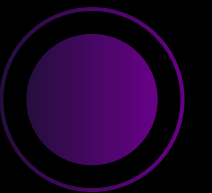
Programming & Scripting: Python, Bash, PowerShell,c/c++.

- **Developping your field-specific skills**

Common Offensive tools and techniques for aspiring red teamers

Assembly and RE for aspiring malware analysts

- **Continious Learning**





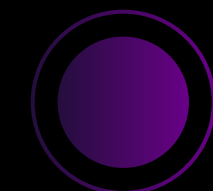
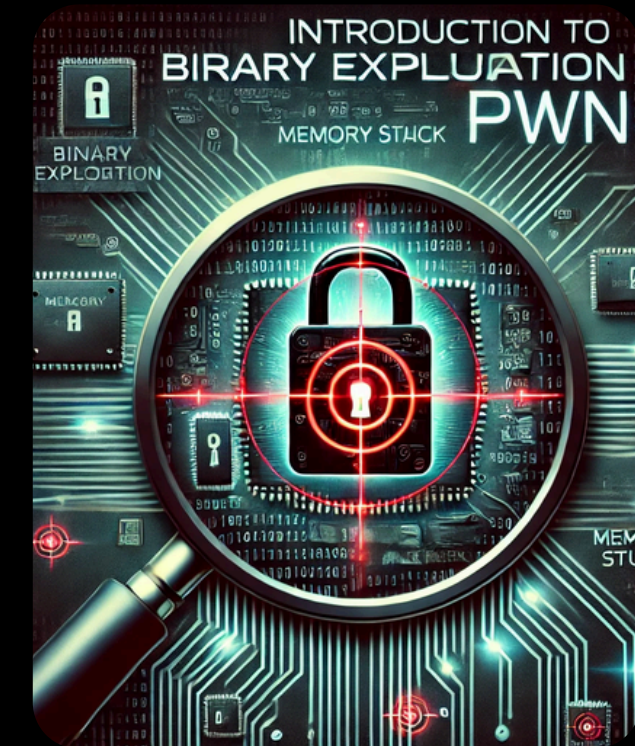
Workshops



Beginner Friendly: Hacking Process, Enumeration, and Recon, Basic RE, Introduction to Web and Systems Architectures

Medium Level: Web Exploitation(CBBH path), Privilege escalation, Buffer overflow exploitation, Malware Analysis

Advanced Level: Exploit Development, Custom payload creation, Malware Development...





Capture The Flag



**nice opinion.
one small issue.**



Forensics



- Memory forensics
- Log Analysis
- File Analysis
- Steganography
- Network Traffic Analysis



Steganography

- Hidden data in images, audio, or video.



WEB EXPLOITATION



Web exploitation is about finding vulnerabilities in programs and utilising them to do what you wish.



What is Web exploitation?





REverse engineering

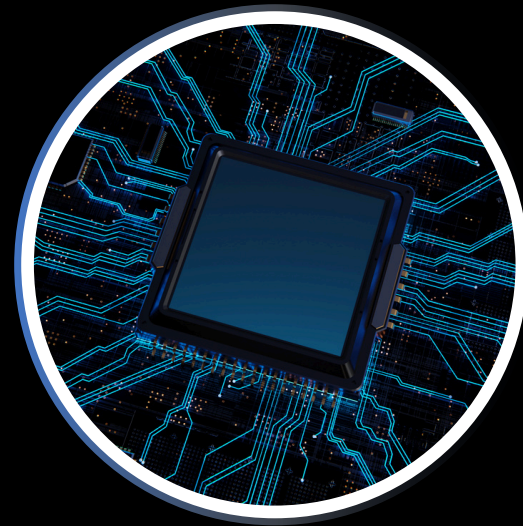
Reverse engineering challenge is to understand the functionality of a given program such that you can identify deeper issues.

🔍 Reverse what? why?





Binary exploitation (Also called pwn)



Vulnerability Identification

Exposing vulnerabilities in the technology used or vulnerabilities in the logic/implementation of a software.



Vulnerability Exploitation

Finding a way in to utilize the vuln to our advantages and make the software do what we order him to do.



WTF is pwn?



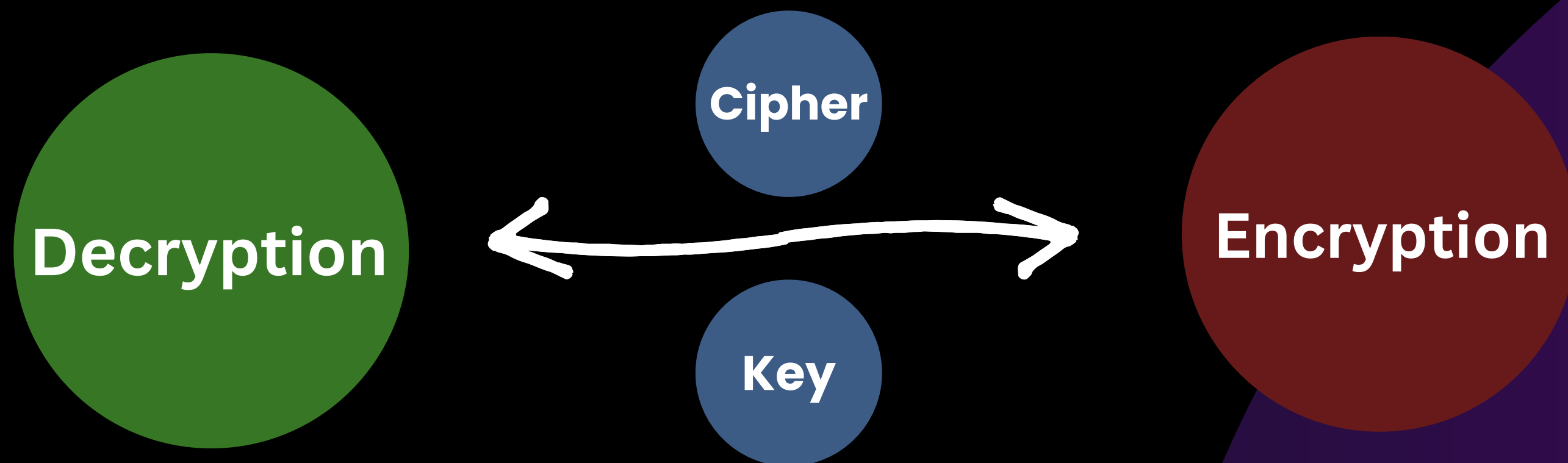
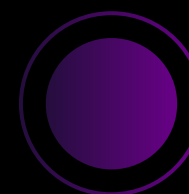
is it difficult?



In a nutshell



Cryptography





Thank You

GET IN TOUCH WITH US

Taha Marweni



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Achraf Kaskoussi



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